



Sarmad Nawaz

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● ABOUT ME

Aspiring bioinformatician skilled in **big data analytics** and **AI-driven biological research**, integrating **computational insights** with advanced **molecular biology techniques**. For my final year project, I am developing a **structured cervical cancer research pipeline** that systematically integrates **ML-based multi-omics biomarker discovery** with **wet-lab validation**.

● EDUCATION AND TRAINING

13/12/2022 – CURRENT Faisalabad, Pakistan

BACHELOR OF SCIENCE IN BIOINFORMATICS (CGPA: 3.30/4.00) Government College University Faisalabad

Website <https://gcuf.edu.pk> | Level in EQF EQF level 7

● PROJECTS

18/09/2025 – 03/01/2026

NeuroScan-AI : MRI Brain Tumor Classification

- Developed **NeuroScan-AI**, a deep learning-based MRI brain tumor classification system, achieving **91.15% accuracy** across four tumor classes.
- Presented the project as an **academic research poster**, demonstrating model design, evaluation, and application of AI for early brain tumor detection.

13/05/2025 – 10/08/2025

Single-cell RNA-seq Analysis Project (PBMCs, 10X Genomics)

- Practiced scRNA-seq skills (Kallisto-BUSTools, Seurat, DropUtils) on PBMC dataset (10X Genomics).
- Performing QC, clustering, UMAP visualization, and marker gene detection.
- Strengthened expertise in cluster annotation (CellAssign, SingleR) and multi-sample integration.

25/02/2025 – 10/05/2025

Bulk RNA-seq Pipeline (Snake-make Workflow)

Developed a **Snakemake-based bulk RNA-seq pipeline** automating raw FASTQ → quantification (Kallisto + tximport) → differential expression (DESeq2/limma) → clustering / PCA → functional enrichment; designed for both single- and paired-end reads and efficient execution on low-spec systems.

17/11/2024 – 19/02/2025

Bulk RNA-seq Analysis Project (Cutaneous Leishmaniasis)

- Practiced RNA-seq skill (QC, Kallisto, DESeq2/limma) on lesion vs. healthy skin samples.
- Identifying isoform-level changes.
- Validated results with ARCHS4 datasets.
- Strengthening skills in DEG detection, DTU analysis, and functional enrichment.

09/07/2024 – 13/11/2024

Xenobiotic-Degrading Protein Prediction Pipeline (Research Project – Attempted)

- Designed a pipeline for **predicting xenobiotic-degrading proteins using machine learning**
- Practiced **pipeline development, model training, and dataset handling**
- Project not completed successfully due to **computational resource limitations**, but provided strong learning experience

16/04/2024 – 15/06/2024

Unique Coding Sequence Project

- Developed a bioinformatics Workflow for identifying unique coding sequences and analyzing genetic features.

● INTERSHIPS

12/08/2025 – 12/11/2025

Wet-Lab Internship – Drug Discovery And Structural Biological Lab, National Institute for Biotechnology and Genetic Engineering (NIBGE), Faisalabad

- Key learnings from this internship are gaining hands-on wet-lab experience in viral **RNA isolation, cDNA synthesis, PCR, and DNA extraction** from **FMDV** (Foot-and-Mouth Disease Virus) samples.
- Working on an ongoing project of **molecular characterization and phylogenetic analysis of newly emerging FMDV**.
- Hands-on experience in **mammalian cell culture, Toxicity assay, RT-PCR, and biosafety level-2 laboratory practice**.

23/06/2025 – 18/07/2025

Summer Internship Bioinformatics, Computational Biology & Data Science – At Integrative Omics and Molecular Modeling Lab, Bioinformatics and Biotechnology Department, Government College And University Faisalabad

- Completed a 4-week internship focused on **Python scripting, programming & scientific computing**
- Hands-on experience with **biological data processing, bioinformatics tools, machine learning & deep learning architectures**
- Worked on **data structures, computational workflows, data visualization & software development practices**
- Practical training on **agentic AI workflow automation using n8n**

● WORKSHOPS AND SEMINARS

09/11/2025 – 09/11/2025

Webinar, From Data to Cure: AI and Bioinformatics in Modern Healthcare

25/08/2025 – 26/08/2025

RNA Sequencing Analysis in Health and Disease (SINES,NUST)

22/02/2025 – 23/02/2025

Hands-on NGS Workshop on Metagenomics for Health and Disease (SINES,NUST)

20/02/2025 – 20/02/2025

Bacterial Identification And Metagenomics (NIBGE)

18/02/2025 – 19/02/2025

From Data To Discovery: Mastering NGS And GWAS (GCUF)

● SKILLS

Dry Lab

Python (computer programming) | Machine Learning | R | Linux | Genomics | Bulk Rna Transcriptomics | Single Cell Transcriptomics | Multi-Omics Integration | Metagenomics | Molecular Docking | Computational Drug Discovery | Pipeline Development

Wet Lab

PCR | RT-PCR | DNA extraction | RNA extraction | Cell Culture | MTT assay | ELISA

Bioinformatics Tools

RStudio | n8n | Github | Kallisto | Kb-python | Bowtie | Biopython | Bioconductor | Docker | Bracken | Kraken | MEGA | Clustal Omega | Uniprot | PHYLIP | Chimera | Ensemble | Blast+ | KEGG | PDB | AutoDock vina | Drugbank | PyRx

Soft Skills

Scientific writing | Project management | Problem-solving | Self-learning

VOLUNTEERING

08/01/2026 – 08/01/2026 Government College University, Faisalabad

Poster Presenter — NeuroScan-AI

- Presented the **NeuroScan-AI** research poster, explaining the deep learning approach and MRI-based brain tumor classification results to faculty and attendees.

12/02/2025 – 12/02/2025 Government College University, Faisalabad

One Student – One Tree Plantation Drive 2025

27/11/2024 – 27/11/2024 Government College University, Faisalabad

One-Day Seminar and Awareness walk on “Climate Change: Smog and Health Impacts”

14/08/2024 – 31/08/2024 Government College University, Faisalabad

Independence Plantation Drive 2024

ACHIEVEMENTS

Sports & Extracurricular Achievements

- Participated in various **division and state-level football championships** representing my city's club team
- Participated in **All-Pakistan Inter-University Football Championship (Higher Education Commission)**
- Reached the **Final Match** in Football Tournament in the **University Annual Sports Tournament**, representing department (Government College University Faisalabad)
- Secured **3rd position** in **Inter-Collegiate Football Championship** at college level (Sahiwal Board)

NETWORKS AND MEMBERSHIPS

Scientific and Social Welfare Societies

• Campus Embassador of OmicsNexus. • Green Youth Movement Club. • Sports Club. • Bioinformatics Society of Pakistan.

RECOMMENDATIONS

Dr. Muhammad Zubair Associate Professor

- My Final Year Project Supervisor at GCUF, from whom I am learning and performing **dry lab predictions** for my final year project.

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DR. Wasim Abbas Principal Scientist

- My Co-Supervisor at NIBGE, from whom I am learning and validating **dry lab predictions in wet lab** for my final year project.

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